



Discrete Variable Conditional 3D Models in Computer Extended Descriptive Geometry (CeDG): Preliminary Study on Cylindrical Polygonal Sheet-Metal Elbow

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1 Introduction

Computer extended Descriptive Geometry (CeDG) technique has demonstrated its ability to build parametric computational models of 3D systems supported by Descriptive Geometry procedures [1]. A CeDG model is constituted by a sequence of graphic-mathematical entities with dependency's relationships between them, which represent the primitives and curves resulting from the projections of the 3D system, as well as the descriptive geometry procedures and other mathematical relations.

This structure allows taking advantage of the available knowledge corpus, as well as extending the scope and accuracy of the associated representation systems, thanks to the properties of the Dynamic Geometry software (GeoGebra) on which CeDG is implemented [2, 3]. As an example, the encounters between spatial surfaces are defined exactly by planar parametric curves representing their projections that are computed by means dependent graphical-mathematical entities and the 2D Locus command [4].

CeDG has some additional advantages of accuracy and scope with respect to current CAD in the construction of flat patterns of developable spatial surfaces, as was shown in a comparative study between CeDG and Solid Edge 2020 [5].

When a parameter causes a qualitative change in the 3D model, it requires additional logic. The treatment of visibility is a well-studied example in CAD that forces the consideration of qualitative (but continuous) changes when varies the projection direction (continuous parameter).

A particularly challenging problem appears with discrete parameters that induce a qualitative and non-continuous change in a 3D model. This work deals with the preliminary study of a methodology for the construction of 3D CeDG models with qualitative changes associated to discrete parameters. We call discrete variable conditional model (DVCM) to this type of 3D CeDG models.

As we discussed in a germinal study [1], the success and strong diffusion of CAD along last decades has displaced the descriptive geometry to a shrinking domain in academic teaching [6]. To the best of our knowledge, CeDG is the first computer-based methodology based on Descriptive Geometry for 3D modelling with focus in both

academic and professional fields. CeDG basis is different to that of CAD, which is based on homogeneous coordinates and approximations surfaces-curves (B-splines and derivatives) from the applied geometry area [7].

A cylindrical polygonal sheet-metal elbow model of n ferrules with the ability to support the change of the discrete variable n ($n > 2$) corresponds to this typology of models. It will be used as a target system to study the scope and accuracy of CeDG (GeoGebra – based) in the construction of discrete variable conditional 3D models. The model is compared against a similar model computed with LogiTRACE v14, and against a Visual Basic based sheet metal elbow model built on CATIA and recently published [8].

2 Methods and Materials

The starting point is a 3D CeDG model of a circumscribed cylindrical polygonal elbow that solves the connection of two circular openings of the same diameter (D), which are located on planes that form an alpha (α) angle. The centres of these openings are placed on a circumference of R_c radius for a fixed number of ferrules (4).

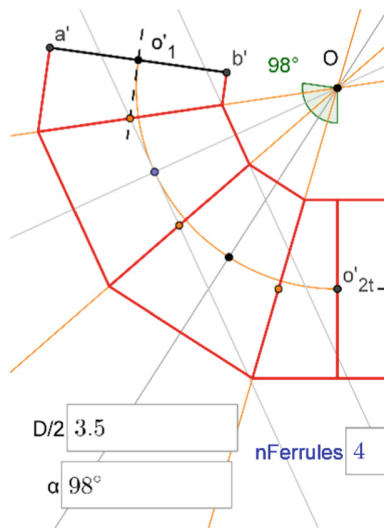


Fig. 1. Vertical projection of the starting 3D CeDG 4 ferrule cylindrical elbow.

Figure 1 shows the vertical projection of this model, where the planes of the openings have been placed perpendicular to the vertical plane and their centres in the same frontal plane. The horizontal projection is not presented in this paper, as its construction is trivial. The model shows how the opening of $o'2t$ centre has been calculated after the original opening has been recessed so that the $o'2t$ and $o'1$ centres lie on a circle of O centre and R_c radius. The values of the model parameters that are shown in Fig. 1 are $D = 70$ mm (Fig. 1 presents it in centimetres), $\alpha = 98^\circ$, and $R_c = 79$ mm. The constructive

dihedral technique described in Izquierdo Asensi [7] was applied, also obtaining the flat pattern of the complete reference cylinder.

In this study, this model has been taken as a reference in order to calculate a DVCM CeDG model that adds the n parameter (number of ferrules, $n\text{Ferrules}$ in the figure) to the parameters already available. For that purpose, it was started with a copy operation with translation of the starting openings, so that they continued keeping the input parameters of the original elbow.

In order to design the discrete variable conditional CeDG model, GeoGebra's GGB-Scripting language has been used. The resultant model was first validated with the fixed four-ferrule CeDG model. Secondly, a cylindrical polygonal model was created with LogiTRACE v14, which is a specific CAD tool for sheet metal systems. This second model was compared with the conditional CeDG model for all n values that are allowed by LogiTRACE, in order to validate the novel DVCM CeDG model. The paper presents the projections for $n = 4$ and 8 and the corresponding flat patterns for $n = 8$.

Results obtained with CeDG are compared through their dimensional values against theoretical expected according to well-known descriptive geometry procedures [9].

3 Results and Discussion

Figure 2 shows the vertical projection of the polygonal cylindrical sheet-metal elbow obtained by the new discrete variable conditional CeDG model, for $n = 4$ and $n = 8$ ferrules. Projections refer to the parameter values defined in the Methods section.

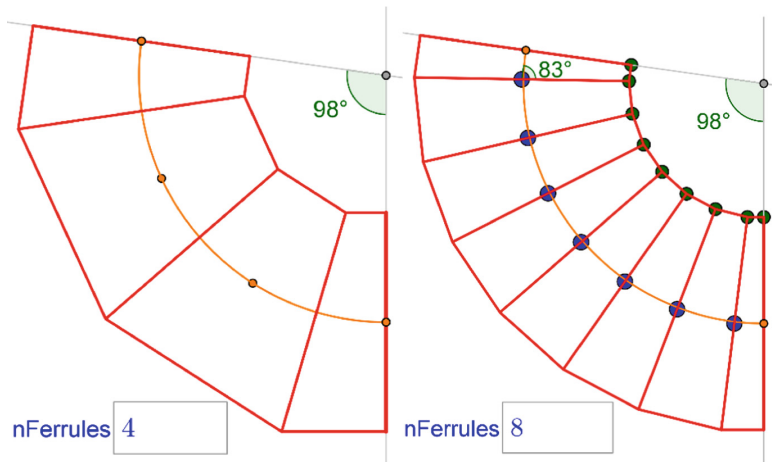


Fig. 2. Vertical projections of the 4 and 8 ferrules cylindrical elbow from the discrete variable conditional CeDG model.

The DVCM CeDG model for $n = 4$ was compared successfully with the standard (fixed n) CeDG model. That is, dimensions of the flat pattern and projections were the same, in agreement with the CeDG methodology, which builds the models through the true geometrical objects.

Figure 3 shows the rendered vertical projection of the sheet-metal elbow for 4 and 8 ferrules obtained by means of LogiTRACE v14, for the same parameter values (and zero plate thickness – neutral fiber).

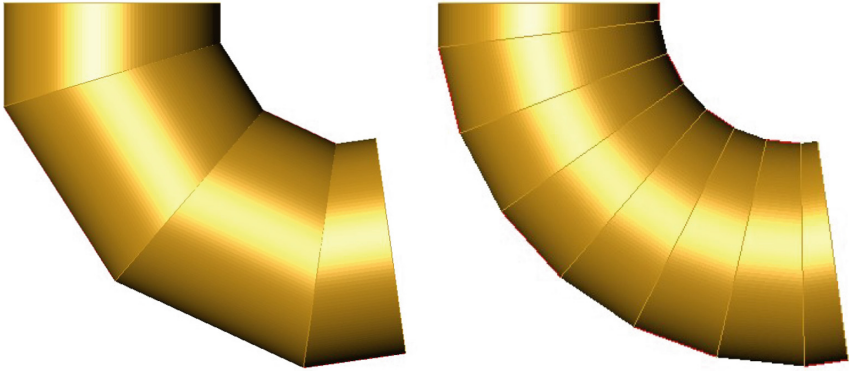


Fig. 3. Vertical (rendered) projections of the 4 and 8 ferrules cylindrical elbow from LogiTRACE v14.

Although we do not know the internal algorithm of the polygonal sheet-metal elbow model in LogiTRACE v14, this software limits the number of ferrules to $n = 12$. When entering a higher value, the software reports that the maximum allowed value has been exceeded and it closes.

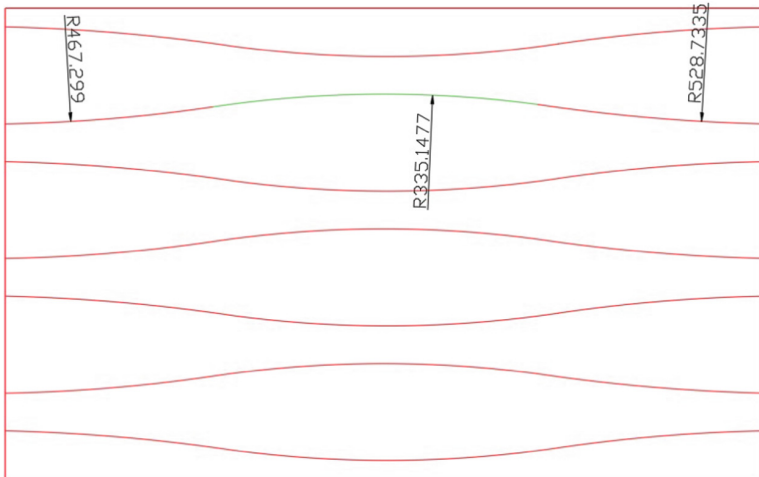


Fig. 4. Flat pattern of the cylindrical elbow for $n = 8$ from LogiTRACE v14 ($NG = 180$).

We have confirmed a very accurate agreement between the DVCM CeDG model and the LogiTRACE model (projections and flat pattern) for the range of $n = 3$ –12 ferrules.

In fact, the flat patterns obtained by CeDG provide the exact values of the transformed curves (for the accuracy that is configured in the software by the number of working decimal places), thanks to the maintenance of the geometrical integrity of the solutions (they are not approximations).

As the flat pattern of logiTRACE is a discrete approximation based on the chosen number of generatrices (NG), a brief inspection of the NG influence on the LogiTRACE solution was performed, using values of $NG = 60$ and 180 . In LogiTRACE, the flat patterns of the connections between ferrules are approximated by three circular arcs, which are presented in Fig. 4 for $NG = 180$. These are 467.29 and 528.73 mm for the arcs that connect to vertical edges (red colour) and 335.15 mm for the central arc (green colour), as it is shown in Fig. 4.

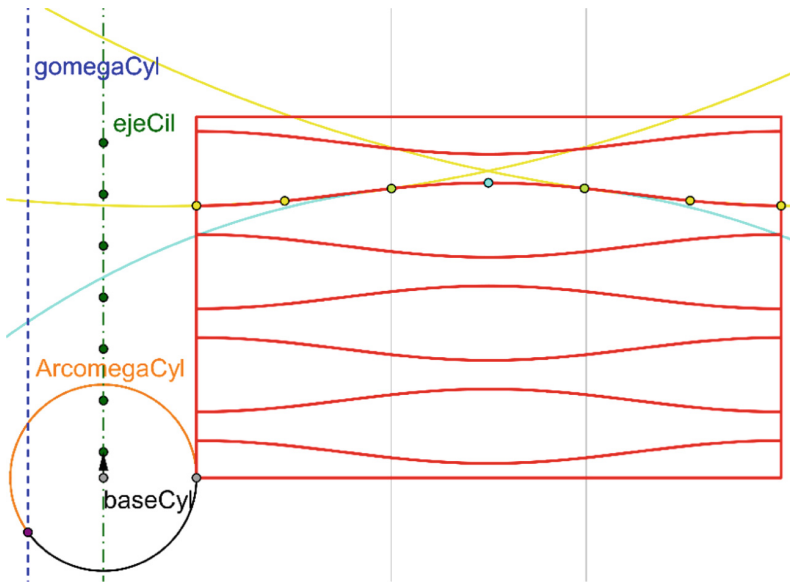


Fig. 5. Flat pattern of the cylindrical elbow for $n = 8$ from the DVCM CeDG model. The figure includes some constructive elements that are addressed in the code presented in the Appendix.

The values of the circular arcs' radii for $NG = 60$ were 461.89 and 528.73 mm for the extreme arcs and 336.06 mm for the central one.

Although this confirms that the influence is not very significant for sufficiently high values of NG, the values of the arc's radii that approximate the exact parametric curve that is calculated by CeDG (which are defined to cover 1/3 of the path each) were 598.26 mm for the extremes and 314.3 mm for the central one. They were obtained by three regular points on the exact transformed curve (see Fig. 5). The radii of the ends are equal by symmetry. This error in the radii (approximately of 10 mm) is due to the influence of approach arc's amplitude.

The DVCM CeDG model of the cylindrical polygonal sheet-metal elbow used an iterative and recurrent logical structure based on Lists objects (type of language object) to

determine the calculation of the sheet metal elbow irrespective of the number of ferrules required. This building technique is quite powerful and provides a computationally efficient starting point for the development of the desired discrete variable conditional modelling methodology in CeDG. The GGBScripting code is presented in the Appendix.

A technique with a lesser scope could have consisted of selecting the sheet metal elbow model corresponding to a fixed number of previously built ferrules but requires having a version of the model for each value of the n parameter. This second solution seems similar to the one recently published in CATIA using Visual Basic code, with an allowed range of ferrules between 2 and 8 [8].

Even though there are standards that limit the number of ferrules to values between 2 and 4 for angles up to 90° (UNE-EN 10224:2003-A1), these are focused on non-alloy steel pipes and fittings for the conduction of aqueous liquids. The ranges of D and R_c are also limited by this standard. Notwithstanding, techniques that allow the building of models with different numbers of ferrules and parameters (physically compatible) are required for sheet metal elbows with angles greater than 90° , different materials and conditions, or for the analysis and research.

4 Conclusions

The study has shown that the novel CeDG technique allows the creation of 3D models that dynamically attend not only to the change of continuous parameters, but also to discrete ones which are associated with qualitative changes of the 3D system. To verify it, we have developed a universal discrete variable conditional model (DVCM), capable of generating the solution for any physically correct value of the number of ferrules.

The developed methodology has proven to be reliable and accurate. This has been verified by contrasting the DVCM CeDG model with a standard 3D CeDG model for a fixed value of n (first step) and with 3D models that were built on LogiTRACE v14 (second step).

This methodology provides significant advantages, including the ability to calculate accurate models (which are not based on approximation curves) of both projections and flat patterns, in contrast to LogiTRACE v14 and CATIA solutions, which are approximations (in the case of flat pattern, they are limited by the number of generatrices that are used). This advantage allows the CeDG solution to be used in engineering and scientific studies that require it [3].

Appendix

The code that implements the DVCM model of the cylindrical elbow is divided into two blocks. The first one obtains the vertical projection of the elbow, defined by planes that are perpendicular to the projection plane, as indicated in the methodology section. If the system were not initially in this position, the model would include the necessary changes of projection direction.

The same list is divided into sub-blocks that define each of the tasks that are performed sequentially. The input data are the ends of the projective segments of the openings. These are assigned to the PortA1, PortA2, PortB1 and PortB2 points handled by the code (blue lines).

Elbow vertical projection

```

# Initialization
Delete(PortA1)
Delete(PortA2)
Delete(PortB1)
Delete(PortB2)
Delete(ListContourA1)
Delete(ListContourA2)
PortA1 = J'
PortA2 = I'
PortB1 = K2
PortB2 = J2
# Execution block - tangent points
PortACenter = Midpoint( PortA1, PortA2 )
PortBCenter = Midpoint( PortB1, PortB2 )
linePortA = Line(PortA1, PortA2)
linePortB = Line(PortB1, PortB2)
CenterArcElbow = Intersect(linePortA, linePortB)
ArcElbow = CircularArc( CenterArcElbow, PortACen-
ter, PortBCenter)
DeltaElbow = Angle(PortA1, CenterArcElbow, PortB1)
DeltaElbowAB = If(DeltaElbow > 180°, -(DeltaElbow-180°),
DeltaElbow)
Ang2dElbow = DeltaElbow / (nFerrules - 1)
ListRadius = IterationList(Rotate(ArcElbowPoint,
Ang2dElbow, CenterArcElbow), ArcElbowPoint, {PortACen-
ter}, nFerrules - 1)

# Consecutive ferrules division planes
ListPlanes = {}; Puntos que definen los planos de corte
[n-1]
ListPlanes = Sequen-
ce(Intersect(Tangent(ListRadius(iFerrules), ArcElbow),
Tangent(ListRadius(iFerrules+1), ArcElbow)), iFerrules,
1, nFerrules - 1); Incluye los tres (n-1) planos de sepa-
ración.
# Contour points
ListContourA1 = {}
SetValue(ListContourA1, 1, PortA1)
ListContourA2 = {}
SetValue(ListContourA2, 1, PortA2)

```

```

iFerrules = 1
linePlanes = Line(CenterArcElbow, ListPlanes(iFerrules))
lineTangent = Tangent(ListRadius(iFerrules), ArcElbow)
DO = {"SetValue(ListContourA1, iFerrules+1, Inter-
sect(linePlanes, Line(ListContourA1(iFerrules), lineTan-
gent)))", "SetValue(ListContourA2, iFerrules+1, Inter-
sect(linePlanes, Line(ListContourA2(iFerrules), lineTan-
gent)))", "SetValue(iFerrules, iFerrules+1)"}
LOOP = {"If(iFerrules<nFerrules,Execute(Join(DO,
LOOP)))"}
Execute(LOOP)
SetValue(ListContourA1, nFerrules+1, PortB1)
SetValue(ListContourA2, nFerrules+1, PortB2)
# Contour segments and planes
iFerrules = 1
Segment(ListContourA1(1), ListContourA2(1))
DO={"Segment(ListContourA1("+ iFerrules +"), ListContou-
rA1("+iFerrules + "+1))", "Segment(ListContourA2("+
iFerrules +"), ListContourA2("+iFerrules + "+1))", "Seg-
ment(ListContourA1(" + iFerrules + "+1), ListContourA2("
+ iFerrules + "+1))", "SetValue(iFerrules, iFerrules+1)"}
LOOP = {"If(iFerrules<=nFerrules,Execute(Join(DO,
LOOP)))"}
Execute(LOOP)

```

The following code block obtains the flat pattern of the ferrules on a reference cylinder. Figure 5 shows the main objects on which the calculation is based. Thus, baseCyl is the real magnitude of the opening, on which the OmegaCyl free point is defined, that identifies the gomegaCyl generatrix and the ArcomegaCyl arc. This figure also shows the vertical projection of the cylinder's axis, axisCil. A list of points (highlighted in green), ListPlanesD, is calculated on the axis. These points represent the intersection with the axis of each plane which, in turn, contains the intersection between two adjacent ferrules.

The angle formed between these planes and the axis (is sign-alternating, which is treated as $(-1)^{iFerrules}$ in the code, DO list) is stored in the PhiPlaneAxis variable (it is shown with the 83° value in Fig. 2) and allows the planes' projections to be defined, their projection points to be calculated and their flat patterns to be finally computed.

Flat pattern

```

# Initialization
Delete(PortACenterD)
Delete(LineBase)
PortACenterD = M_2
LineBase = t;
ejeCil = PerpendicularLine(PortACenterD, LineBase)
vEje = UnitVector(Vector(PortACenterD, (x(PortACenterD),
y(PortACenterD)+1)))
dEje = Distance(ListPlanes(1), ListPlanes(2))
# Rectangle
PBL = (x(PortACenterD) + Distance(PortA1, PortA2)/2 ,
y(PortACenterD))
PBR = (x(PBL) +  $\pi$  Distance(PortA1, PortA2),
y(PortACenterD))
PTL = (x(PBL) , y(PortACenterD) + (nFerrules-1) dEje)
PTR = (x(PBR), y(PTL))
Segment(PBL, PBR)
Segment(PTL, PTR)
Segment(PBL, PTL)
Segment(PBR, PTR)
# Axis - Planes intersection in Pattern
ListPlanesD = IterationList(PEje + dEje vEje, PEje,
{PortACenterD + dEje/2 vEje}, nFerrules - 2)
PhiPlaneAxis = Angle(ListContourA2(2), ListPlanes(1),
PortACenter)
baseCyl = Circle(PortACenterD, PBL)
omegaCyl = Point(baseCyl)
gomegaCyl = Line(omegaCyl, ejeCil)
# Main loop with transformed connections
ArcomegaCyl = Arc(baseCyl, PBL, omegaCyl)
iFerrules = 1
DO = {"Locus(Intersect(y = y(Intersect(Line(ListPlanesD(
"+iFerrules+"), Rotate(PortACenterD, (-1)^" +
iFerrules+"*PhiPlaneAxis, ListPlanesD(" + iFerrules +
"))), gomegaCyl)), x = x(PBL)+Length(ArcomegaCyl)),
omegaCyl)", "SetValue(iFerrules, iFerrules+1)"}
LOOP = {"If(iFerrules<=nFerrules-1, Execute(Join(DO,
LOOP)))"}
Execute(LOOP)

```

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